

# PHISHING EMAIL INVESTIGATION

## INVESTIGATION REPORT

SOC Analyst Portfolio Project | Email Threat Analysis

| Field               | Details                           |
|---------------------|-----------------------------------|
| Analyst             | Meenakshy B R                     |
| Date                | April 2025                        |
| Case ID             | PHI-2025-001                      |
| Severity            | HIGH                              |
| Attack Type         | Phishing Email                    |
| Target              | phishing@pot                      |
| Sender              | banco.bradesco@atendimento.com.br |
| Impersonated Entity | Banco do Bradesco Livelo (Brazil) |
| Verdict             | CONFIRMED PHISHING                |

### 1. EXECUTIVE SUMMARY

A phishing email sample impersonating Banco do Bradesco Livelo, a major Brazilian bank, was analyzed as part of a SOC analyst investigation exercise. The email falsely claimed the recipient had 92,990 reward points expiring immediately — a classic urgency-based social engineering tactic designed to pressure victims into clicking malicious links.

Technical analysis confirmed the email failed ALL standard email authentication checks (SPF, DKIM, DMARC) and was sent from a cheap Linux VPS server hosted on DigitalOcean — not from any legitimate Bradesco banking infrastructure. The sender domain (atendimento.com.br) has no affiliation with the official Bradesco Bank domain (bradesco.com.br).

**Verdict: CONFIRMED PHISHING. The email exhibits multiple technical and behavioral indicators of a sophisticated phishing campaign targeting Brazilian banking customers.**

### 2. EMAIL DETAILS

#### 2.1 Basic Email Information

| Field               | Value                    | Suspicious?               |
|---------------------|--------------------------|---------------------------|
| Sender Display Name | BANCO DO BRADESCO LIVELO | Looks legitimate but FAKE |

|                  |                                                                                      |                                |
|------------------|--------------------------------------------------------------------------------------|--------------------------------|
| Sender Email     | banco.bradesco@atendimento.com.br                                                    | 🚩 Not official Bradesco domain |
| Official Domain  | bradesco.com.br                                                                      | Completely different domain!   |
| Recipient        | phishing@pot                                                                         | 🚩 Not a real person            |
| Subject          | CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje! | 🚩 Urgency tactic               |
| Date Sent        | Tue, 19 Sep 2023 18:35:49 UTC                                                        |                                |
| Return Path      | root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06                                           | 🚩 Linux VPS server!            |
| Content Encoding | base64                                                                               | 🚩 Used to hide malicious links |

### 3. TECHNICAL ANALYSIS

#### 3.1 Email Authentication Analysis (MXToolbox)

All three standard email authentication checks failed for this email:

| Check    | Result                     | What It Means                                             |
|----------|----------------------------|-----------------------------------------------------------|
| SPF      | ❌ FAILED                   | Email cannot prove it came from a legitimate server       |
| DKIM     | ❌ NONE — Not Signed        | Email has NO digital signature — banks always sign emails |
| DMARC    | ❌ FAILED — No Record Found | Domain has zero email protection policy                   |
| CompAuth | ❌ FAILED (reason=001)      | Microsoft flagged this as suspicious                      |

All three authentication checks failing simultaneously is a definitive indicator of email spoofing and phishing. Legitimate financial institutions always pass these checks.

#### 3.2 Email Routing Analysis

The email relay path reveals the true origin of this email:

- Origin Server: ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06 — a cheap rented Linux VPS
- Origin IP: 137.184.34.4 — belongs to DigitalOcean LLC (ASN 14061)
- Location: United States — a Brazilian bank email sent from a US cloud server
- 55 second delay at Hop 2 — Microsoft spam filter flagged and held the email
- Hop 2 and Hop 5 marked with errors — Microsoft's system detected anomalies

No legitimate bank sends emails from a cheap Ubuntu Linux VPS server. This clearly indicates a threat actor rented a disposable server to send phishing emails in bulk.

### 3.3 IP Address Investigation (VirusTotal)

| Field            | Finding                                                                                           |
|------------------|---------------------------------------------------------------------------------------------------|
| IP Address       | 137.184.34.4                                                                                      |
| Owner            | DigitalOcean LLC                                                                                  |
| ASN              | 14061                                                                                             |
| Location         | United States                                                                                     |
| Vendor Flags     | 0/94 — Not yet blacklisted                                                                        |
| alphaMountain.ai | Not Recommended  |

Although the IP was not widely blacklisted, this is a common characteristic of freshly provisioned phishing servers. Threat actors regularly rent new cloud servers specifically because they have no negative reputation yet. The use of DigitalOcean infrastructure to impersonate a bank is a significant red flag.

### 3.4 Domain Investigation (Whois)

| Field                    | Finding                                 |
|--------------------------|-----------------------------------------|
| Domain                   | atendimento.com.br                      |
| Official Bradesco Domain | bradesco.com.br — completely different! |
| Domain Owner             | Maximilian Gregory Peisker Lacerda      |
| Created                  | 2018-09-20                              |
| Host                     | Hostgator Brazil                        |
| Affiliation to Bradesco  | NONE — completely unrelated domain      |

The sender domain atendimento.com.br has absolutely no affiliation with Banco do Bradesco. The word 'atendimento' means 'service' or 'support' in Portuguese — a deliberate choice to appear legitimate to Brazilian banking customers. This domain was likely compromised or used as a spoofed sender identity.

## 4. INDICATORS OF COMPROMISE (IOCs)

| Type          | Value                             | Description                        |
|---------------|-----------------------------------|------------------------------------|
| Sender Email  | banco.bradesco@atendimento.com.br | Fake Bradesco sender address       |
| Sender Domain | atendimento.com.br                | Unaffiliated spoofed domain        |
| Origin IP     | 137.184.34.4                      | DigitalOcean VPS — phishing server |

|                  |                                       |                                 |
|------------------|---------------------------------------|---------------------------------|
| Sending Server   | ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06 | Linux VPS — not a bank server   |
| ISP Abused       | DigitalOcean LLC (ASN 14061)          | Cloud hosting used for phishing |
| Authentication   | SPF=fail, DKIM=none, DMARC=fail       | All checks failed               |
| Content Encoding | base64                                | Used to hide malicious content  |

## 5. RED FLAGS IDENTIFIED

- Sender domain (atendimento.com.br) is NOT the official Bradesco domain (bradesco.com.br)
- SPF authentication FAILED — email cannot prove legitimate origin
- DKIM = None — email has NO digital signature whatsoever
- DMARC FAILED — Microsoft's compauth marked it as suspicious (reason=001)
- Email sent from a Linux Ubuntu VPS server — not from any banking infrastructure
- Origin IP belongs to DigitalOcean — commonly abused for phishing campaigns
- Extreme urgency language — 'points expiring TODAY!' — pressure tactic
- Email body encoded in base64 — used to hide malicious links from scanners
- alphaMountain.ai flagged origin IP as 'Not Recommended'
- 55 second delay at Microsoft spam filter — email was flagged on arrival
- Return-Path shows Linux server hostname — not a bank mail server
- Generic recipient address (phishing@pot) — mass phishing campaign indicator

## 6. MITRE ATT&CK MAPPING

| Technique ID | Technique Name           | Description                                |
|--------------|--------------------------|--------------------------------------------|
| T1566        | Phishing                 | Main attack technique — deceptive email    |
| T1566.001    | Spearphishing Attachment | Email with malicious content               |
| T1598        | Phishing for Information | Credential harvesting via fake bank portal |
| T1036        | Masquerading             | Impersonating legitimate Bradesco bank     |
| T1071.003    | Web Protocols            | Using email as attack delivery mechanism   |

## 7. CONCLUSION

This email is a confirmed phishing attempt targeting Brazilian banking customers. The threat actor carefully crafted the email to impersonate Banco do Bradesco Livelô using a convincing display name and a domain that appears related to banking services.

However, technical analysis reveals multiple definitive indicators of phishing — failed email authentication across SPF, DKIM, and DMARC, origination from a disposable DigitalOcean VPS server, use of base64 encoding to hide malicious content, and a sender domain completely unaffiliated with Bradesco Bank.

**The urgency-based subject line about expiring reward points is a classic social engineering technique designed to bypass the victim's critical thinking and prompt immediate action.**

## 8. RECOMMENDATIONS

### Immediate Actions

- Block sender domain atendimento.com.br at email gateway
- Block origin IP 137.184.34.4 at firewall level
- Delete this email from all employee mailboxes immediately
- Report the phishing domain to PhishTank and abuse@digitalocean.com
- Alert all employees about this phishing campaign

### Long-term Preventive Measures

- Enforce SPF, DKIM, and DMARC policies on company email domain
- Deploy email security gateway with phishing detection
- Conduct regular phishing awareness training for all employees
- Implement email sandboxing to analyze suspicious attachments safely
- Enable multi-factor authentication on all employee accounts
- Create a simple process for employees to report suspicious emails

## 9. EVIDENCE REFERENCES

| # | File                        | Description                               |
|---|-----------------------------|-------------------------------------------|
| 1 | virustotal-ip-analysis.png  | VirusTotal scan of origin IP 137.184.34.4 |
| 2 | whois-domain-lookup.png     | Whois lookup of atendimento.com.br        |
| 3 | mxtoolbox-auth-failures.png | SPF, DKIM, DMARC all failed               |
| 4 | mxtoolbox-relay-path.png    | Email routing showing Linux VPS origin    |
| 5 | mxtoolbox-headers.png       | Full email header analysis                |